POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT V: Non-Conventional Power and Nuclear Power Station

1. Solar Energy – Introduction

- The Sun is a gaseous sphere (diameter 1.39 x 10^6 km, surface temp. ~ 5800 K); energy from ***nuclear fusion*** of hydrogen into helium.
- ***Solar constant*** I_{sc} = ***1367 W/m²*** (energy per unit time on unit area perpendicular to rays, at mean earth-sun distance, outside the atmosphere). About ***1000 W/m²*** reaches the ground on a clear day.
- Types of radiation: ***beam (direct)***, ***diffuse*** (scattered by atmosphere) and ***global (total) = beam + diffuse***; reflected (albedo).
- India receives ***4–7 kWh/m²/day*** with 250–300 clear days -> a huge potential (~ 750 GW). Instruments: ***pyranometer*** (global / diffuse), ***pyrheliometer*** (beam), sunshine recorder (Campbell-Stokes).

Utilisation of solar energy

- ***Thermal route:*** solar water heater, dryer, cooker, still, ***solar thermal power*** (concentrating), space heating, refrigeration / air conditioning.
- ***Photovoltaic route:*** solar cells -> DC electricity (rooftop, solar parks, pumps, lighting).
- Advantages: free, clean, abundant, no moving parts. Limits: ***intermittent*** (night, clouds), low density (needs large area), storage cost, high initial cost.

2. Solar Collectors – Working Principle

- A ***solar collector*** is a special heat exchanger that absorbs solar radiation and transfers the heat to a fluid (water, air, oil).

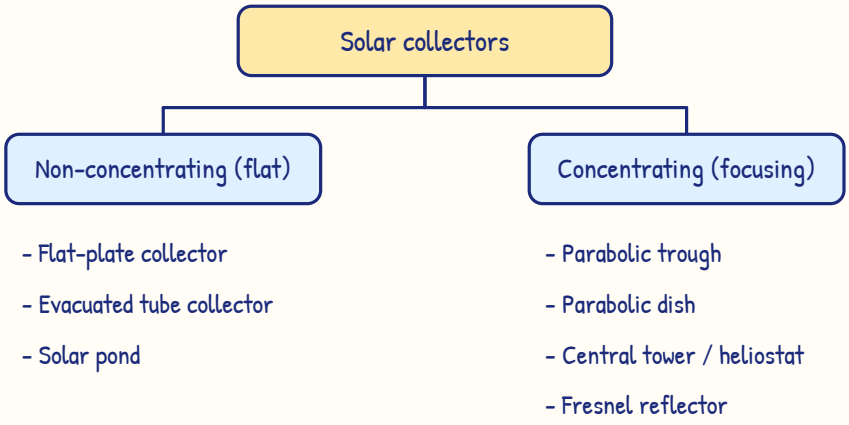


Fig: Classification of solar collectors

Flat-plate collector (FPC)

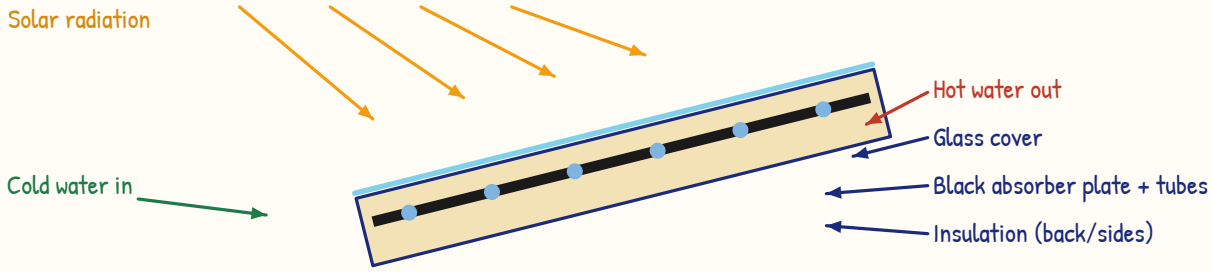


Fig: Flat-plate collector (cross-section)

- **Parts:** (1) **Black absorber plate** (copper / aluminium, selective coating -> high absorptivity, low emissivity); (2) **Tubes (risers + headers)** bonded to the plate carrying water; (3) **Transparent cover** (1 or 2 glass sheets) -- passes short-wave solar radiation but blocks the long-wave (**greenhouse effect**) and cuts convection loss; (4) **Insulation** (glass wool / PUF) at back and sides; (5) **Casing (box)**.
- **Working:** solar radiation passes the glass -> absorbed by plate -> heat conducted to tubes -> water heated (temperature 40-80 deg C). Natural (**thermosyphon**) circulation with tank above, or forced circulation with a pump.

$$\text{Useful heat } Q_u = A_c F_R [S - U_L (T_i - T_a)];$$
$$\text{Efficiency} = Q_u / (A_c I_T)$$

- S = absorbed radiation, U_L = overall heat-loss coefficient, F_R = heat removal factor, T_i = inlet fluid temperature, T_a = ambient temperature. **Tilt = latitude** facing south (in India).

Flat-plate	Concentrating
Uses beam + diffuse radiation	Only beam radiation; needs **sun tracking**
Temp. up to ~ 100 deg C	Temp. 150–1000 deg C
Low cost, simple, little maintenance	Costly; complex
Concentration ratio = 1	Concentration ratio 10–1000+

Concentrating collectors

- **Parabolic trough:** long curved mirror focuses sun on a receiver tube (oil, 400 deg C) -> steam -> turbine (SEGS plants). **Parabolic dish:** point focus, Stirling engine. **Central receiver (power tower):** field of **heliostats** reflects onto the tower boiler (up to 1000 deg C; molten salt storage).
- **Concentration ratio (C)** = aperture area / receiver area.
- **Other uses:** solar still (1 L/m^2/day), **solar pond** (salinity-gradient), solar cooker, **solar PV**: p-n junction, efficiency 15–22%; storage in batteries.

Example 1: Flat plate collector

- Collector area 2 m^2, incident radiation 800 W/m^2, efficiency 55%. Water flow 0.02 kg/s (c_p = 4.18 kJ/kg K).
- Useful heat = 0.55 x 800 x 2 = **880 W**. Temperature rise = 880 / (0.02 x 4180) = **10.5 deg C**.

3. Wind Energy

- Winds are caused by **uneven solar heating** of the earth (about 1–2% of solar energy converts to wind). Wind turbines convert the **kinetic energy of moving air** into rotation -> electrical energy.

$$\text{Power in wind } P = (1/2)\ \rho\ A\ V^3\ (W);\ \text{Power extracted} = (1/2)\ \rho\ A\ V^3\ C_p$$

- rho = air density (1.225 kg/m^3), A = swept area = pi D^2/4, V = wind speed. **Power depends on the cube of wind speed** -> site selection is critical.

- **Betz limit:** maximum possible power coefficient $C_{p,max} = 16/27 = 0.593$ (59.3%). Practical C_p = 0.35–0.45.
- **Tip speed ratio (TSR)** = blade tip speed / wind speed = omega R / V (4–8 for 3-blade); **cut-in speed** ~ 3–4 m/s; **rated speed** 12–15 m/s; **cut-out speed** ~ 25 m/s (turbine shuts down for safety).

Example 2

- Wind speed 12 m/s, rotor diameter 80 m, C_p = 0.4. Swept area = pi x 40^2 = 5027 m^2.
- P = 0.5 x 1.225 x 5027 x 12^3 x 0.4 = **2.13×10^6 W = 2.1 MW**.

Types of wind turbines

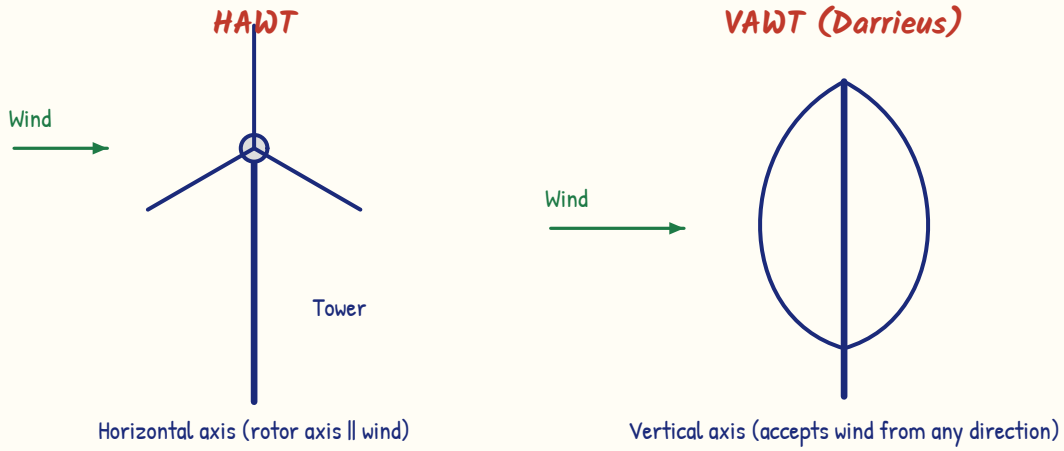


Fig: Horizontal-axis and vertical-axis wind turbines

Feature	HAWT	VAWT
Axis of rotation	Horizontal (parallel to wind)	Vertical (perpendicular to wind)
Yaw mechanism	Needed (turn into wind)	Not needed; accepts any direction
Efficiency	Higher (C_p 0.4–0.5)	Lower (0.3–0.4)
Types	1, 2, **3-blade** (standard), multi-blade (water pumping)	**Savonius** (drag type), **Darrieus** (lift type), H-rotor
Gearbox / generator	Placed at the top of tower (nacelle) -> maintenance difficulty	At ground level -> easy maintenance

Feature	HAWT	VAWT
Tower	Tall (80–120 m), needs strong wind	Short; suitable for urban / low wind
Self-starting	Yes	**Savonius yes; Darrieus no**
Usage	Large grid-connected wind farms	Small stand-alone, rooftop

Components of a HAWT

- **Rotor blades** (GRP / carbon fibre, aerofoil), **hub**, **pitch** system (blade angle control), **nacelle** containing low-speed shaft, **gearbox**, high-speed shaft, **brake**, **generator** (DFIG / PMSG / induction), **yaw drive**, **anemometer + wind vane**, **controller**, **tower**, foundation, transformer.
- **Control:** pitch control and stall control; yaw to face the wind; **variable-speed** operation with power electronics converters.
- **Wind farm:** array of turbines, spacing 5–10 D downwind and 3–5 D crosswind (wake effects); **offshore** wind has higher, steady wind.
- India: **~ 45 GW** (Tamil Nadu, Gujarat, Karnataka, Rajasthan, Maharashtra); Muppandal and Jaisalmer. Advantages: free, no emission, quick installation. Disadvantages: intermittent, noise, bird hazards, visual impact, site-specific, grid variability.

4. Tidal Energy

- Tides are the rise and fall of sea level caused by **gravitational attraction of the moon (and sun)** on the oceans. Two high and two low tides every day (period 12 h 25 min). **Spring tides** (new / full moon) -> large range; **neap tides** small range.
- Useful where the **tidal range > 5 m** and the shape of basin allows enclosing (e.g., Bay of Fundy, La Rance – France 240 MW, Sihwa – Korea; India: **Gulf of Khambhat, Gulf of Kutch, Sundarbans**).

Energy per tide $E = (1/2)\ \rho\ g\ A\ R^2$ (J); Average power $P = E / T$ (T = tide period 12.4 h)

- A = basin area, R = tidal range, rho = 1025 kg/m^3 (sea water).

Example 3

- Basin area 10 km^2 (10^7 m^2), range 6 m. E = 0.5 x 1025 x 9.81 x 10^7 x 36 = ***1.81 x 10^{12} J = 503 MWh*** per filling. If only ~ 80% efficiency and generation is used over 6 h, average power ~ ***70 MW***.

Types of tidal power plants

- ***Single-basin, single-effect:*** one basin; fills through sluices at high tide, ***generates at ebb*** only (4-5 h/day).
- ***Single-basin, double-effect:*** generates on both flood (inflow) and ebb (outflow) using ***reversible turbines***.
- ***Double-basin (two-pool):*** upper and lower basins give ***continuous / controlled*** output; costlier.
- ***Tidal stream / current turbines:*** like underwater wind turbines (no barrage, less environmental impact), tidal lagoons.
- Turbines: ***bulb***, ***rim (Straflo)*** and tubular types -- low head, axial flow; sluice gates; embankment; power house.
- Advantages: renewable, predictable, no fuel. Disadvantages: high cost, intermittent, low load factor (~ 25-30%), site-limited, disrupts marine life, corrosion of equipment, effect on silting.
- ***Ocean wave and OTEC*** (ocean thermal energy conversion using 20 deg C temperature difference between surface and deep water) are other ocean sources.

5. MHD Power Generation

- ***Magneto-Hydro-Dynamics:*** an electrically conducting ***fluid (ionised gas / plasma)*** moving across a ***magnetic field*** generates electricity directly (Faraday's law) -- ***no moving mechanical parts***.

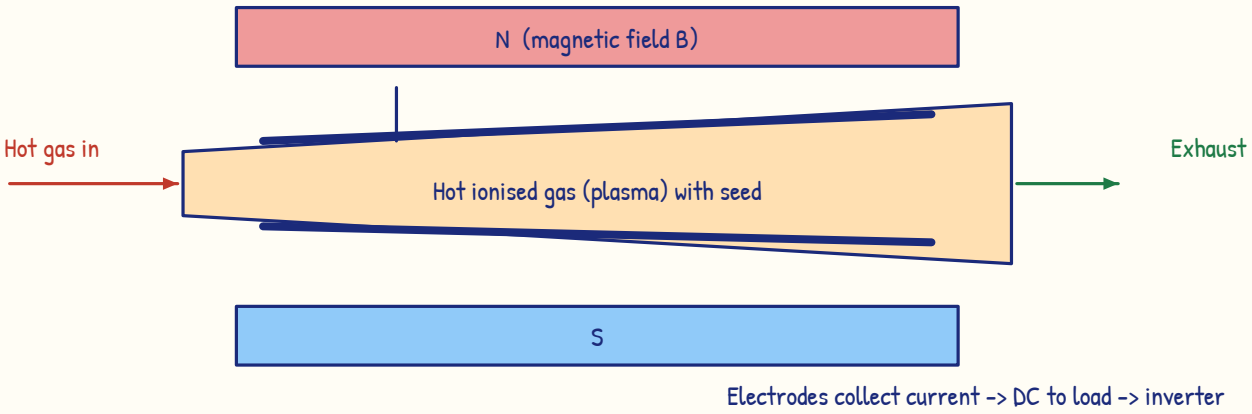


Fig: Principle of an open-cycle MHD generator

> *Induced emf $E = B\ v\ L$ (V), Power density = sigma $v^2\ B^2$ (W/m^3) (maximum at load factor 0.5)*

- Conductor: hot **ionised gas** from burning fuel at ~ 2500–3000 K, seeded with **potassium or caesium** (easy to ionise) to raise conductivity. The B field is produced by **superconducting magnets**; electrodes collect DC current.

Types

- **Open cycle:** fuel (coal / oil / gas) burned with oxygen-enriched / preheated air, seeded, expanded through the nozzle into channel, then exhaust passes through **air pre-heater, seed recovery**, and a steam bottoming cycle.
- **Closed cycle:** inert gas (helium / argon) + seed (caesium) circulates in a loop heated by a heat source (nuclear reactor, ~ 1800–2000 K). Liquid metal MHD (two-phase) also studied.

Advantages / disadvantages

- **Efficiency 50–60%** in **combined MHD-steam** plant (MHD alone 20%); quick start, no moving parts, less pollution, uses coal.
- Difficult materials (very high temp., electrode erosion), superconducting magnet cost, seed recovery, low conductivity; MHD is still at a **pilot stage** (e.g., in the U.S., Russia U-25, India).

6. Nuclear Power Station – Nuclear Fuel

- **Nuclear power plant:** heat released by **nuclear fission** in a reactor -> steam -> turbine -> generator (Rankine cycle). Only the heat source differs from a steam plant.
- **Mass defect and energy:** $E = \Delta m c^2$; 1 kg U-235 gives about 8.2×10^{13} J (= 2.5×10^6 kg coal); each fission releases ~ **200 MeV**.

Nuclear fuels

- **Fissile (directly fissionable by thermal neutrons):** U-235 (0.7% in natural uranium), **Pu-239, U-233**.
- **Fertile (converted into fissile by neutron capture): U-238 -> Pu-239, Th-232 -> U-233**.
- Natural uranium (0.72% U-235) used in **PHWR**; **enriched** uranium (2–5% U-235) in light-water reactors; fuel as **UO_2 pellets** in zirconium-alloy (zircaloy) clad tubes, bundled into fuel assemblies. Mining -> milling (yellowcake) -> conversion -> enrichment -> fabrication.

7. Nuclear Fission and Chain Reaction

- **Fission:** a heavy nucleus (U-235) absorbs a **slow (thermal, 0.025 eV) neutron** and splits into **two lighter fragments** (e.g., Ba-141 and Kr-92) + **2–3 neutrons** + energy (~ 200 MeV, mostly kinetic energy of fragments -> heat) and gamma rays.

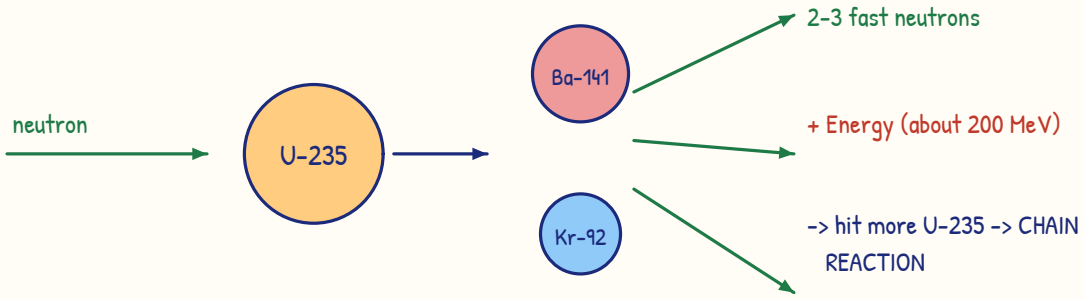


Fig: Fission of U-235 and release of neutrons

- U-235 + n -> Ba-141 + Kr-92 + 3 n + about 200 MeV (mass numbers: 235 + 1 = 141 + 92 + 3)

Chain reaction

- Neutrons released cause further fissions -> **self-sustaining chain reaction**. Measured by the **multiplication factor k = neutrons in one generation / neutrons in the previous generation**.

k value	State	Meaning
k < 1	**Sub-critical**	reaction dies out
k = 1	**Critical**	steady power (normal operation)
k > 1	**Super-critical**	power rises (start-up; bomb if uncontrolled)

- **Critical mass** = minimum fuel mass to sustain a chain reaction. Neutrons are lost by **leakage** and **non-fission capture**; the **four-factor formula** k_{inf} = eta epsilon p f; with leakage $k_{eff} = k_{inf} \times P_{NL}$.
- **Delayed neutrons (0.65%)** give time (seconds) to control the reactor mechanically.

Moderator

- Slows fast neutrons (2 MeV) to thermal (0.025 eV) energy by collisions: **light water, heavy water (D_2O), graphite, beryllium**. Good moderator = low atomic mass, high scattering and low absorption. Heavy water is best (allows natural uranium).

8. Breeding and Fertile Materials

- **Breeding** = producing more fissile material than is consumed. **Fertile materials** (U-238, Th-232) absorb neutrons and decay to fissile nuclei.
- **U-238 + n -> U-239 -> (beta decay) Np-239 -> (beta) Pu-239** (fissile). **Th-232 + n -> Th-233 -> Pa-233 -> U-233** (fissile).
- **Conversion ratio** = fissile atoms produced / fissile atoms consumed; if > 1 it is called **breeding ratio (BR)**.
- **Breeder reactor:** fast neutrons (no moderator), liquid sodium coolant, a core of Pu-239 surrounded by a **blanket of U-238**. **Fast breeder reactor (FBR)** BR ~ 1.2–1.4 -> converts the abundant U-238 into fuel ->

uses 60–70 x more energy from uranium. Doubling time 10–30 years.

- ***India's three-stage programme:*** Stage 1: ***PHWR*** (natural U) -> Pu. Stage 2: ***FBR*** (Pu + U-238; PFBR Kalpakkam 500 MW) -> U-233. Stage 3: ***Thorium-U-233 reactors*** (AHWR) -- uses India's huge thorium reserves.

9. Nuclear Reactor – Components and Types

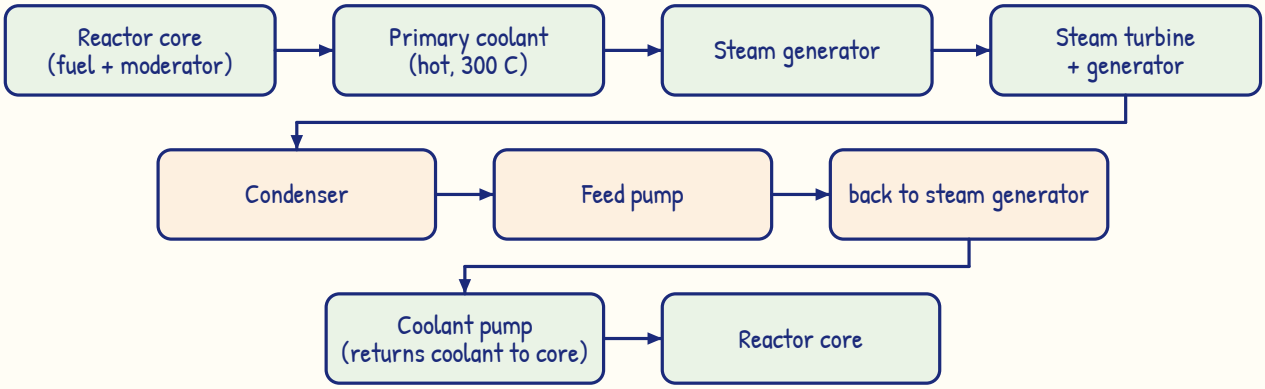


Fig: Block diagram of a nuclear power plant (PWR)

Main components of a reactor

- ***Fuel*** (UO_2 rods), ***moderator***, ***control rods*** (boron, cadmium, hafnium: absorb neutrons; ***shut-down / safety / regulating***), ***coolant*** (water, heavy water, CO_2, helium, liquid sodium), ***reflector*** (graphite / beryllium to reduce leakage), ***shielding*** (concrete + steel: biological shield), ***reactor pressure vessel***, ***containment building***.
- ***Coolant requirements:*** high heat capacity, low neutron absorption, non-corrosive, stable.

Reactor	Fuel	Moderator	Coolant	Features
PWR	Enriched UO_2	Light water	Light water at 150 bar (no boiling)	Most common; steam generator; secondary loop
BWR	Enriched UO_2	Light water	Light water boils in core (70 bar)	Direct steam to turbine; simpler; radioactive steam

Reactor	Fuel	Moderator	Coolant	Features
PHWR (CANDU)	Natural UO_2	Heavy water	Heavy water	On-load refuelling; used in India
GCR / AGR	Natural / enriched	Graphite	CO_2	High temperature
FBR	Pu + U	None (fast neutrons)	Liquid sodium	Breeds fuel
HTGR	Coated particle	Graphite	Helium	800–950 deg C; hydrogen production

- Indian stations: **Tarapur** (BWR 1969), **Rawatbhata**, **Kalpakkam** (PHWR, PFBR), **Narora**, **Kakrapar**, **Kaiga**, **Kudankulam** (VVER-1000 PWR, Russian), Gorakhpur. Capacity ~ 8 GW.

10. Reactor Operation

- **Start-up:** control rods withdrawn slowly -> neutron flux rises (monitored by **neutron detectors** -> source range -> intermediate -> power range) until k = 1 (critical) -> power raised to the desired level and turbine coupled.
- **Power control:** inserting control rods reduces k (and power); withdrawing raises power. Also **boric acid** (soluble poison in PWR) for slow control; coolant flow and temperature feedback.
- **Shutdown:** all rods (**SCRAM**) inserted automatically on abnormal conditions (high power, high temperature, low flow) within seconds. After shut-down, the **decay heat** (~ 7% initially) still needs cooling -> **emergency core cooling system (ECCS)**, passive cooling.
- **Self regulating features: negative temperature coefficient** and void coefficient make the reactor stable.
- **Poisons:** Xenon-135 and Samarium-149 absorb neutrons -> affect restart soon after shut-down (**xenon pit**).
- **Fuel burn-up** ~ 7000 MWd/t (PHWR) to 40,000–60,000 MWd/t (PWR). Refuelling: on power (CANDU) or shut-down (PWR, every 12–18 months). Spent fuel -> **cooling pond** -> reprocessing (**PUREX**) -> waste

vitrification and geological disposal.

Safety and radiation

- **Defence in depth:** fuel clad -> pressure boundary -> **containment** (reinforced concrete). Dose limits: public 1 mSv/year; radiation workers 20 mSv/year (averaged). **AERB** regulates in India.
- Accidents: Three Mile Island (1979), **Chernobyl (1986)**, **Fukushima (2011)** -> improved passive safety.

Advantages of nuclear power	Disadvantages
Tiny fuel quantity; very high energy density	High capital cost; long construction
No CO_2 / SO_2 / ash; reliable **base load** (CF > 80%)	Radioactive waste (long-lived); decommissioning
Large reserves with breeders / thorium	Accident risk, public acceptance, security
Low fuel cost	Needs skilled staff, cooling water; thermal pollution; poor load following

Example 4: Fuel consumption

- A 300 MW (electrical) plant, efficiency 30%, so thermal power = 1000 MW. Energy per fission 200 MeV = 3.2×10^{-11} J.
- Fissions per second = $10^9 / 3.2 \times 10^{-11}$ = **3.1×10^{19} /s**. Mass of U-235 consumed per day = $3.1 \times 10^{19} \times 86400 \times 235 \times 1.66 \times 10^{-27}$ kg = **~ 1.05 kg / day**.
- Compare coal: 1000 MW(t) needs about **3000+ tonnes of coal per day**.

Q & A (exam revision)

- **What is the solar constant?** 1367 W/m^2 on a surface normal to the sun's rays outside the earth's atmosphere.
- **Why is glass cover used in a flat-plate collector?** It transmits short-wave solar radiation but traps the long-wave heat from the plate (greenhouse effect) and reduces convective loss.

- **Why does a wind turbine have a cut-out speed?** To protect the rotor, gearbox and tower from damage in storms (about 25 m/s).
- **Why seeding in MHD?** Gas at 2500–3000 K alone has low conductivity; alkali metal (K, Cs) ionises easily and raises conductivity.
- **Role of moderator vs control rod?** Moderator slows neutrons to thermal energy; control rods absorb neutrons to regulate k.
- **Why heavy water for natural uranium?** Heavy water absorbs very few neutrons and moderates well, so enough neutrons remain to sustain the chain reaction without enrichment.
- **What is a SCRAM?** Rapid automatic insertion of all control rods to shut down the reactor.

Quick revision

- Solar constant 1367 W/m^2. FPC: black plate + glass cover + insulation. Wind power = 0.5 rho A V^3; Betz limit 59.3%.
- Tidal energy = 0.5 rho g A R^2; need range > 5 m. MHD: conducting gas + magnetic field -> direct DC; 50–60% in combined cycle.
- Fission: U-235 + n -> fragments + 2–3 n + 200 MeV. k = 1 critical. Moderator slows, control rods absorb, coolant carries heat.
- Fertile: U-238, Th-232; fissile: U-235, Pu-239, U-233. FBR breeds fuel (BR > 1).